3. 设 $M=(a_{ij})_{m\times n}$ $M^*=(b_{ij})_{n\times m}$

$A\vec{v_i}=\sum_{l=1}^{m} a_{li}\vec{w_l}$

$A^*\vec{w_j}=\sum_{k=1}^{n} b_{kj}\vec{v_k}$

$\langle A^*\vec{w_j},\vec{v_i}\rangle=\langle\vec{w_j},A\vec{v_i}\rangle$

即 $\sum_{k=1}^{n} b_{kj}\langle\vec{v_k},\vec{v_i}\rangle=\sum_{l=1}^{m} a_{li}\langle\vec{w_j},\vec{w_l}\rangle$

$$[\langle v_i,v_1\rangle,\langle v_i,v_2\rangle\cdots\langle v_i,v_n\rangle]\begin{bmatrix}b_{1j}\\ \vdots\\ b_{nj}\end{bmatrix}=[\langle w_j,w_1\rangle\cdots\langle w_j,w_m\rangle]\begin{bmatrix}a_{1i}\\ \vdots\\ a_{mi}\end{bmatrix}$$

即 $(G_v\cdot M^*)_{ij}=(G_w,M)_{ji}$

$G_v\cdot M^*=[G_w M]^T=M^T\cdot G_w^T=M^T\cdot G_w$

$M^*=G_v^{-1}\cdot M^T\cdot G_w$

$w_j,w\rangle_W$

4. (1) $\forall\vec{v_2}=A^*(w)\in R(A^*)$

$\langle\vec{v_1},\vec{v_2}\rangle_V=\langle\vec{v_1},A^*(\vec{w})\rangle_V=\langle A(\vec{v_1}),\vec{w}\rangle_W=\langle 0,\vec{w}\rangle_W=0$

$\therefore$ $\forall\vec{v_1}\in Ker(A)$, $\forall\vec{v_2}\in R(A^*)$, $\vec{v_1}\perp\vec{v_2}$.

(2) 设 $\forall\vec{v}\in V$, $\vec{v_1}\in Ker(A)$, 令 $\vec{v_2}=\vec{v}-\vec{v_1}$

下证 $\vec{v_2}\in R(A^*)$:

$\because$ $\vec{v_1}\in Ker(A)$ $\therefore$ $A\vec{v_1}=0$

$\vec{v}=\vec{v_1}+\vec{v_2}$

$A(\vec{v})=A(\vec{v_1})+A(\vec{v_2})$ $\therefore$ $A\vec{v}=A\vec{v_2}$ $A\vec{v_2}\in R(A)\in W(W^*)$

~~若 $\vec{v_2}\perp R(A^*)$~~

~~即对 $\forall\vec{w}\in W$, $\langle\vec{v_2},A^*(\vec{w})\rangle=\langle A\vec{v_2},\vec{w}\rangle=0$ 则 $A\vec{v_2}=0$~~

~~$\therefore A\vec{v}=0$ 则 $\vec{v}\in Ker(A)$ $\therefore$ $\vec{v_2}=0$, 这与 $\vec{v_2}=\vec{v}-\vec{v_1}$ 矛盾~~ 矛盾!

由 Riesz 表示定理 $f_{Av_2}(\vec{w})=\langle Av_2,\vec{w}\rangle=\langle v_2,A^*\vec{w}\rangle$

又 $A^*\vec{w}\in R(A^*)\subseteq V(V^*)$ $\therefore$ 上式 $=f_{v_2}(A^*w)$ $\therefore v_2\in R(A^*)$.

唯一性: 假设存在两对 v_1,v_2, v_1',v_2' 使 $v=v_1+v_2=v_1'+v_2'$

$v_1-v_1'=v_2-v_2'$

$v_1-v_1'\in Ker(A)$ $v_2-v_2'\in R(A^*)$

但 $R(A^*)={}^{\perp}Ker(A)$

$\langle v_1-v_1',v_2-v_2'\rangle=0$ $\therefore$ $v_1-v_1'=0=v_2-v_2'$ $\therefore\begin{cases}v_1=v_1'\\ v_2=v_2'\end{cases}$ 矛盾.

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5.(1) Gram矩阵 $G=\begin{pmatrix} \langle v_1,v_1\rangle, & \langle v_1,v_2\rangle & \langle v_1,v_3\rangle \\ \langle v_2,v_1\rangle, & \langle v_2,v_2\rangle, & \langle v_2,v_3\rangle \\ \langle v_3,v_1\rangle, & \langle v_3,v_2\rangle & \langle v_3,v_3\rangle \end{pmatrix}$

$$=\begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$$

(2)
$$\begin{pmatrix} x \\ y \\ z \end{pmatrix}=\lambda\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}+\mu\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}+z\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

$$\begin{cases} \mu+z=x \\ \lambda+z=y \\ \lambda+\mu=z \end{cases} \quad \begin{cases} \lambda=\dfrac{-x+y+z}{2} \\ \mu=\dfrac{x-y+z}{2} \\ z=\dfrac{x+y-z}{2} \end{cases}$$

其 ( ) 表示以 $\begin{pmatrix}1\\0\\0\end{pmatrix}\begin{pmatrix}0\\1\\0\end{pmatrix}\begin{pmatrix}0\\0\\1\end{pmatrix}$ 为基底的坐标，[ ] 表示以 $\vec{v_1},\vec{v_2},\vec{v_3}$ 为基底的坐标

$$\left\langle \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}, \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix} \right\rangle = \left\langle \begin{bmatrix} \frac{-x_1+y_1+z_1}{2} \\ \frac{x_1-y_1+z_1}{2} \\ \frac{x_1+y_1-z_1}{2} \end{bmatrix}, \begin{bmatrix} \frac{-x_2+y_2+z_2}{2} \\ \frac{x_2-y_2+z_2}{2} \\ \frac{x_2+y_2-z_2}{2} \end{bmatrix} \right\rangle =$$

$$=\left\langle \left(\frac{-x_1+y_1+z_1}{2}\right)\vec{v_1}+\left(\frac{x_1-y_1+z_1}{2}\right)\vec{v_2}+\left(\frac{x_1+y_1-z_1}{2}\right)\vec{v_3},\ \left(\frac{-x_2+y_2+z_2}{2}\right)\vec{v_1}+\left(\frac{x_2-y_2+z_2}{2}\right)\vec{v_2}+\left(\frac{x_2+y_2-z_2}{2}\right)\vec{v_3} \right\rangle$$

$$=\frac{(-x_1+y_1+z_1)(-x_2+y_2+z_2)}{4}+\frac{(x_1-y_1+z_1)(x_2-y_2+z_2)}{4}+\frac{(x_1+y_1-z_1)(x_2+y_2-z_2)}{4}$$

(3) $G'=\begin{pmatrix} \langle\vec{e_1},\vec{e_1}\rangle, & \langle\vec{e_1},\vec{e_2}\rangle, & \langle\vec{e_1},\vec{e_3}\rangle \\ \langle\vec{e_2},\vec{e_1}\rangle, & \langle\vec{e_2},\vec{e_2}\rangle, & \langle\vec{e_2},\vec{e_3}\rangle \\ \langle\vec{e_3},\vec{e_1}\rangle, & \langle\vec{e_3},\vec{e_2}\rangle, & \langle\vec{e_3},\vec{e_3}\rangle \end{pmatrix}=\begin{pmatrix} \frac{3}{4}, & -\frac{1}{4}, & -\frac{1}{4} \\ -\frac{1}{4}, & \frac{3}{4}, & -\frac{1}{4} \\ -\frac{1}{4}, & -\frac{1}{4}, & \frac{3}{4} \end{pmatrix}$

(4) $\|\vec{e_1}\|=\sqrt{\langle\vec{e_1},\vec{e_1}\rangle}=\frac{\sqrt{3}}{2}$

$\|\vec{e_2}\|=\sqrt{\langle\vec{e_2},\vec{e_2}\rangle}=\frac{\sqrt{3}}{2}$

设夹角为θ

$$\cos\theta=\frac{\langle\vec{e_1},\vec{e_2}\rangle}{\|\vec{e_1}\|\cdot\|\vec{e_2}\|}=\frac{\frac{1}{4}}{\frac{3}{4}}=\frac{1}{3}$$

$\theta=\arccos\frac{1}{3}$，$\theta\in\left(0,\frac{\pi}{2}\right)$

$S=\|\vec{e_1}\|\cdot\|\vec{e_2}\|\sin\theta=\frac{\sqrt{3}}{2}\cdot\frac{\sqrt{3}}{2}\cdot\frac{2\sqrt{2}}{3}=\frac{\sqrt{2}}{2}$

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6. $\Leftarrow$ 当Gram矩阵可逆时

若线性相关 则∃一组不全为0的$a_1 \dots a_n$. 使得

$a_1\vec{v_1} + \dots + a_n\vec{v_n} = \vec{0}$ 两边与$\vec{v_i}$作内积,

$a_1\langle\vec{v_1}\cdot\vec{v_i}\rangle + a_2\langle\vec{v_2}\cdot\vec{v_i}\rangle + \dots + a_n\langle\vec{v_n}\cdot\vec{v_i}\rangle = 0$.

即 $a_1\langle\vec{v_i},\vec{v_1}\rangle + \dots + a_n\langle\vec{v_i},\vec{v_n}\rangle = 0$

$G\vec{a} = \vec{0}$. $\vec{a} = (a_1, a_2 \dots a_n)^T$

∵ G是可逆的 ∴ $\vec{a}$只有零解 矛盾! 所以线性无关

$\Rightarrow$ 当线性无关时

则 $a_1\vec{v_1} + \dots + a_n\vec{v_n} = \vec{0} \Leftrightarrow \vec{a} = (a_1 \dots a_n)^T = \vec{0}$

左右两边同时作$\vec{v_i}$内积

$a_1\langle\vec{v_i},\vec{v_1}\rangle + \dots + a_n\langle v_i, v_n\rangle = 0$.

$G\vec{a} = \vec{0}$ 若不可逆，则∃非零解$\vec{a}$. $G\vec{a} = \vec{0}$

矛盾! ∴ 可逆